UNIT III

MATHEMATICS FOR DATA SCIENCE

Matrix Algebra • Random Vectors • Principal Component Analysis

Lecture Notes, Step-by-Step Worked Examples, Practice Questions, and Exam Preparation

Study guide • formulas • numerical methods • applications

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1. Vectors and Matrices

A vector is an ordered list of numbers. A matrix is a rectangular array of numbers. In a common data-matrix convention, rows represent observations and columns represent variables.

```
x = [2, 3, 4]^T
```

Matrix addition requires matrices of the same dimensions. Matrix multiplication AB is defined when the number of columns of A equals the number of rows of B. In general, AB is not equal to BA.

Worked example: matrix multiplication

Let A = [[1, 2], [3, 4]] and B = [[2, 0], [1, 3]]. Multiply each row of A by each column of B.

```
(AB)11 = 1×2 + 2×1 = 4; (AB)12 = 1×0 + 2×3 = 6
```

```
(AB)21 = 3×2 + 4×1 = 10; (AB)22 = 3×0 + 4×3 = 12
```

```
AB = [[4, 6], [10, 12]]
```

Why used: matrix multiplication represents linear transformations and is central to covariance calculations and PCA.

2. Positive Definite Matrices

A real symmetric matrix A is positive definite if x^T A x > 0 for every nonzero vector x. It is positive semidefinite if x^T A x >= 0 for every vector x.

- Eigenvalue test: for symmetric A, all eigenvalues positive implies positive definite.
- Sylvester's criterion: all leading principal minors must be positive.
- For A = [[a,b],[b,c]], positive definite iff a > 0 and ac – b² > 0.
- Uses: optimization, covariance analysis, and quadratic forms.

Worked example

For A = [[4,1],[1,3]], the first leading principal minor is D1 = 4 > 0. The determinant is D2 = 4×3 – 1×1 = 11 > 0. Both are positive, so A is positive definite.

3. Matrix Square Root

A square-root matrix B of A satisfies B² = A. If A = QΛQ^T is real symmetric positive semidefinite, its principal square root is A^(1/2) = QΛ^(1/2)Q^T, taking the nonnegative square root of each eigenvalue.

Worked example

```
A = [[4,0],[0,9]] ⇒ A^(1/2) = [[2,0],[0,3]]
```

```
Verification: [[2,0],[0,3]]² = [[4,0],[0,9]] = A
```

Why used: covariance transformations and multivariate statistics.

4. Random Vectors and Matrices

A random vector is a vector of random variables, X = [X1, X2, …, Xp]^T. Its expectation is calculated componentwise: E[X] = [E(X1), E(X2), …, E(Xp)]^T. For a random matrix, calculate the expectation of each entry.

```
If E(X1)=25 and E(X2)=70, then E[X]=[25,70]^T
```

Why used: representing uncertainty in multivariate data.

5. Mean Vectors

The mean vector contains the mean of each variable. Suppose X = [[2,4],[4,6],[6,8]], with observations in rows.

```
Mean of variable 1 = (2+4+6)/3 = 4

Mean of variable 2 = (4+6+8)/3 = 6

Mean vector x■ = [4, 6]^T
```

Why used: describes the central location of multivariate data and is needed to center observations before PCA.

6. Covariance Matrices

Population covariance: Cov(X,Y) = E[(X–μX)(Y–μY)]. Sample covariance: sXY = Σ(xi–x■)(yi–■)/(n–1). Use n for population covariance and n–1 for the usual sample covariance.

```
For centered data Xc, sample covariance matrix S = Xc^T Xc / (n–1)
```

Diagonal entries are variances; off-diagonal entries are covariances. Positive covariance means variables tend to move together; negative covariance means they tend to move in opposite directions. Zero covariance does not necessarily mean independence.

Worked example: covariance matrix

Given X = [[1,2],[2,4],[3,6]], the mean vector is [2,4]^T.

```
Centered Xc = [[–1,–2],[0,0],[1,2]]

Xc^T Xc = [[2,4],[4,8]]

S = (1/(3–1)) Xc^T Xc = [[1,2],[2,4]]
```

Interpretation: the sample variances are 1 and 4, and the sample covariance is 2. This matrix is the usual starting point for covariance-based PCA.

7. Quadratic Forms and Matrix Inequalities

A quadratic form is Q(x) = x^T A x. The Rayleigh quotient is R(x) = (x^T A x)/(x^T x), for x ≠ 0. For real symmetric A, λmin ≤ R(x) ≤ λmax.

Under x^T x = 1, the maximum of x^T A x equals the largest eigenvalue, attained by a corresponding unit eigenvector.

Worked example: maximization

Let A = [[3,0],[0,1]], with x^T x = 1. The eigenvalues are 3 and 1. Therefore max x^T A x = 3, attained at x = [1,0]^T or its negative.

Why used: this optimization result underlies the first PCA direction.

8. Principal Component Analysis (PCA)

PCA transforms correlated variables into uncorrelated linear combinations called principal components. PC1 captures the largest possible variance among unit-length linear combinations. PC2 captures the largest remaining variance subject to orthogonality to PC1.

- Uses: dimensionality reduction, visualization, noise reduction, feature extraction, and multivariate quality monitoring.
- PCA preserves variance, not necessarily predictive power for a particular target.
- Center data first; standardize when scales differ and correlation-based PCA is appropriate.

9. Population Principal Components

```
Population covariance: Σ = E[(X–μ)(X–μ)^T]
For a unit vector a, Var(a^T X) = a^T Σ a
PC1 solves max a^T Σ a subject to a^T a = 1
```

The solution is an eigenvector of Σ corresponding to its largest eigenvalue. Subsequent principal components correspond to descending eigenvalues and orthogonal eigenvectors. The variance of PCi equals λi.

Worked example

Let Σ = [[4,0],[0,2]]. The characteristic equation is |Σ–λI| = (4–λ)(2–λ) = 0, so λ1=4 and λ2=2.

```
v1 = [1,0]^T; v2 = [0,1]^T
PC1 = X1, variance 4; PC2 = X2, variance 2
PC1 variance percentage = 4/(4+2) × 100% = 66.67%
```

10. Summarizing Sample Variation

```
Total variance = Σ λi
Percentage explained by PCi = λi / (Σj λj) × 100%
Cumulative percentage for first k PCs = (Σi=1..k λi / Σj λj) × 100%
```

Worked example

For eigenvalues 6, 3, 1, total variance is 10. PC1 explains 60%, PC2 explains 30%, and PC3 explains 10%. The first two components explain 90% cumulatively. If at least 90% is required, two components suffice for this example.

11. PCA Scores and Graphing

```
If Vk contains retained eigenvectors as columns, scores are Z = Xc Vk
```

For centered x = [2,1]^T, let v1 = [1,1]^T/√2 and v2 = [1,–1]^T/√2.

```
PC1 = v1^T x = 3/√2 ≈ 2.121
PC2 = v2^T x = 1/√2 ≈ 0.707
```

Plot each observation at (PC1 score, PC2 score). Score plots help inspect clusters, outliers, and major directions of variation. Eigenvectors must come from the dataset's covariance or correlation matrix.

12. Large-Sample Inference

```
Sample mean vector: x■ = (1/n) Σi Xi
Under standard assumptions: E[x■] = μ; Cov(x■) = Σ/n
```

Worked example

If n=100 and Σ = [[100,20],[20,25]], then:

```
Cov(x■) = Σ/100 = [[1,0.2],[0.2,0.25]]
```

Increasing sample size generally decreases the sampling variability of the sample mean vector.

13. Quality Monitoring with PCA

PCA can summarize many process measurements into a smaller number of component scores. A typical procedure is:

- Collect historical measurements from a stable process.
- Center variables; standardize if appropriate.
- Calculate covariance or correlation matrix.
- Find eigenvalues/eigenvectors and calculate component scores.
- Monitor component scores and residual variation with validated control statistics.
- Investigate observations that exceed established control limits.

Monitoring may use Hotelling's T^2 and a residual statistic such as Q. Limits must be established for the specific process; an alarm signals investigation, not automatic proof of a fault.

14. Exam Question Patterns

High-priority numerical questions

- Find a mean vector from a data matrix.
- Calculate a sample covariance matrix step by step.
- Determine whether a symmetric matrix is positive definite.
- Find a principal square root of a matrix.
- Find eigenvalues/eigenvectors and principal components.
- Calculate individual and cumulative percentages of variance explained.
- Maximize x^T A x subject to x^T x = 1.
- Calculate PCA scores and interpret a score plot.
- Calculate Cov(x■) = Σ/n.
- Explain PCA-based quality monitoring.

Theory questions

- Define random vector, random matrix, mean vector and covariance matrix.
- Explain positive definite and positive semidefinite matrices.
- Define a matrix square root and principal components.
- Explain the role of eigenvalues and eigenvectors in PCA.
- Explain how PCs summarize sample variation and how scores are graphed.
- Describe large-sample inference for a mean vector.
- Describe PCA-based quality monitoring.

Questions may ask for direct calculation, verification, interpretation, constrained optimization, a real-world application, or a combined multi-step PCA analysis.

15. Complete PCA Workflow for Exams

- 1. Arrange observations in rows and variables in columns, unless stated otherwise.
- 2. Calculate each column mean and form the mean vector.
- 3. Center the data: Xc = X – 1 x■^T.
- 4. Calculate sample covariance: S = Xc^T Xc/(n–1).
- 5. Solve |S–λI| = 0 for eigenvalues.
- 6. Find and normalize eigenvectors; order by descending eigenvalues.

- 7. Calculate individual and cumulative variance percentages.
- 8. Choose the number of components using the stated criterion.
- 9. If requested, calculate scores Z = Xc Vk and interpret or plot them.

16. Practice Questions with Step-by-Step Answers

Question 1: Find the mean vector

Given X = [[2,4],[4,6],[6,8]].

```
x■1 = (2+4+6)/3 = 4; x■2 = (4+6+8)/3 = 6
```

```
Answer: x■ = [4,6]^T
```

Question 2: Test positive definiteness

A = [[5,1],[1,2]] is symmetric. D1 = 5 > 0; det(A) = 5×2 – 1×1 = 9 > 0. Both leading principal minors are positive, so A is positive definite.

Question 3: Find the principal square root

```
A = diag(16,25) ⇒ A^(1/2) = diag(4,5)
```

Check: diag(4,5)² = diag(16,25).

Question 4: Percentage of variance explained

Eigenvalues are 8, 1, 1. Total variance = 8+1+1 = 10.

```
PC1 percentage = (8/10)×100% = 80%
```

Question 5: Maximize a quadratic form

For A = diag(5,2), subject to x^T x = 1, eigenvalues are 5 and 2. The maximum is the largest eigenvalue.

```
Maximum = 5, attained at x = [1,0]^T or its negative.
```

17. Exam Revision Priorities

Priority	Topics to prepare
Very high	Covariance matrices; PCA eigenvalues/eigenvectors; variance explained; positive definiteness; quadratic-form maximi
High	Mean vectors; matrix square roots; PCA scores and graphing; PCA-based quality monitoring
Medium–high	Large-sample inference and theory definitions

These are study priorities based on topic connections, not verified examination weightage. In numerical answers, show the formula, substitutions, intermediate calculations, interpretation, and a boxed conclusion.